Activation vs. Valence in Noise: Diagnosing the Phonetic Vulnerabilities of an SER System

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Abstract

Speech Emotion Recognition (SER) systems remain vulnerable to acoustic noise, limiting real-world deployment. This study systematically investigates performance degradation in phonetic-acoustic terms, examining how additive white Gaussian noise (AWGN) disrupts the phonetic cues underlying emotional expression. An XGBoost classifier trained on RAVDESS (768 files, 24 speakers, and 4 emotions: angry, calm, happy, sad) was evaluated across Signal-to-Noise Ratios (SNRs) from 20dB to -10dB.

Results revealed catastrophic failure: overall accuracy declined from 75% (clean) to 25% (-10dB SNR), with critical degradation at ~5dB SNR, typical café noise levels. Analysis through the lens of phonetic dimensions (activation vs. valence) exposed a fundamental asymmetry in robustness. The model successfully maintained activation distinctions (high vs. low energy) even under severe noise, but valence discrimination collapsed entirely.

Feature importance analysis confirmed over-reliance on activation-correlated cues: four of the top five features measured signal energy (`stdEnergy`, `maxEnergy`, `mfcc0_mean`, `meanEnergy`). The model's primary failure mode was activation-based confusion, distinguishing anger from happiness (both high-activation) or sadness from calmness (both low-activation) became impossible once AWGN masked the subtle spectral dynamics (delta-delta MFCCs) encoding valence. Counterintuitively, 'calm' proved most robust to noise (F1: 0.56 at -10dB), suggesting its stable, speaker-independent phonetic signature (low F0, low RMS energy) provides a more reliable acoustic template than the dynamic modulations of other emotions.

Phonetic Basis of Emotional Speech

Emotional expression modulates speech production through two primary phonetic mechanisms, corresponding to the psychological dimensions of *activation* (high energy vs. low energy) and *valence* (positive vs. negative affect):

- **Activation correlates:** Fundamental frequency (F0) mean and range, intensity (RMS energy), speech rate, and articulatory precision. High-activation emotions (anger, joy) exhibit elevated F0, increased loudness, faster tempo, and hyper articulation. Low-activation states (sadness, calmness) show suppressed F0, reduced energy, slower tempo, and hypoarticulation.
- **Valence correlates:** Voice quality (breathiness, tenseness), spectral tilt, formant trajectories, and dynamic micro-prosodic variation. Positive valence (happiness, calmness) often involves brighter timbre and smoother

contours, while negative valence (anger, sadness) shows tense voice quality or irregular phonation and steeper spectral slopes.

Crucially, activation cues are *robust*, they manifest as large-scale, stable modulations in amplitude and periodicity that survive acoustic degradation. Valence cues, however, are *fragile* encoded in subtle spectral shape changes and dynamic trajectory variations that broadband noise readily masks. This phonetic asymmetry predicts the observed failure pattern.

Key Findings

1. Catastrophic Degradation at Typical Noise Levels

Performance collapsed from 75% accuracy (clean) to 48% (20dB), 43% (5dB), and 25% (-10dB). The 5dB threshold, equivalent to a busy café or quiet office, represents a critical failure point for real-world deployment. At -10dB (construction noise), the system performs at chance level.

2. Phonetic Diagnosis: Activation Preserved, Valence Lost

Confusion matrices reveal systematic arousal-based grouping under noise. The model-maintained ability to distinguish high-activation (angry, happy) from low-activation (sad, calm) categories, but within these groups, valence discrimination failed catastrophically:

- At -5dB SNR: 17/32 angry samples misclassified as happy (same activation, opposite valence); 22/32 sad samples misclassified as calm (same activation, opposite valence)
- This pattern confirms that AWGN selectively destroys valence-encoding phonetic features while preserving gross activation cues

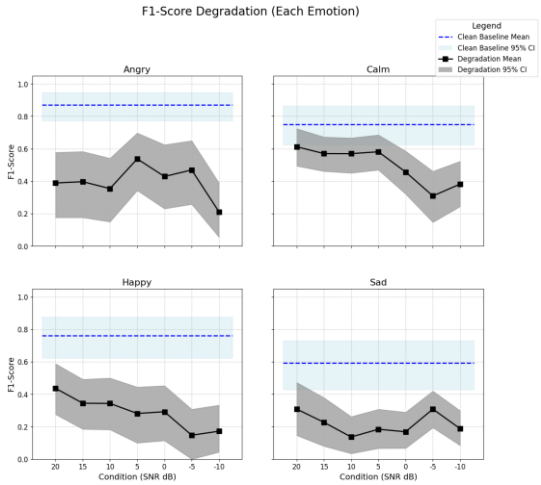

Figure 1: Per Emotion F1-Score Degradation across SNR levels

'Calm' maintains highest robustness due to its stable, low-energy phonetic signature, while high-activation emotions show severe vulnerability as noise masks dynamic spectral cues.

3. Feature Vulnerability: Energy vs. Spectral Dynamics

XGBoost feature importance (weight metric) exposed extreme reliance on activation:

- **Top 5 features (67% total importance):** `stdEnergy` (27%), `maxEnergy` (21%), `mfcc0_mean` (20%, correlates with log-energy), `meanEnergy` (18%), `delta2_mfcc7_mean` (21%)
- **Remaining features:** Delta-delta MFCCs (spectral acceleration), phonetically interpretable as dynamic formant transitions and spectral tilt changes, crucial for valence but highly noise-sensitive

The asymmetry is phonetically expected: RMS energy and F0-correlated features (mfcc0) reflect gross vocal tract output and source characteristics, robust global properties. Delta-delta MFCCs capture rapid spectral shape changes from articulatory gestures and voice quality shifts, precisely the fragile, localized cues AWGN obliterates first.

4. Counterintuitive Robustness of 'Calm'

Despite being a low-energy, phonetically 'weak' category, 'calm' achieved the highest noise-robustness (F1-score: 0.56 at -10dB, vs. 0.17-0.22 for other emotions). Phonetic explanation: 'calm' exhibits a highly *stable* and *speaker-independent* phonetic signature, consistently low F0, suppressed energy, reduced F0 range, creating a canonical acoustic template. Other emotions require fine-grained spectral discrimination between speakers' idiosyncratic realizations, making them vulnerable when noise corrupts these details.

Implications for Phonetically Informed SER

This study demonstrates that current SER architectures fail not due to insufficient data or modelling capacity, but because they exploit the *wrong phonetic cues*, over-relying on robust-but-insufficient activation features while neglecting fragile-but-essential valence markers. For real-world robustness:

- **De-emphasise energy-based features:** Reduce model dependency on global amplitude measures; prioritize spectral shape and dynamic trajectory features
- **Develop valence-specific robust representations:** Extract noise-invariant correlates of voice quality (e.g., H1-H2, cepstral peak prominence) and formant dynamics resistant to broadband interference
- **Leverage speaker-independent canonical patterns:** The success of 'calm' suggests focusing on emotions with stable, low-variance phonetic profiles rather than complex, speaker-dependent realizations
- **Multi-condition training with phonetic awareness:** Train on noise-augmented data with explicit supervision on valence-encoding features to force models to learn noise-resistant discriminative patterns

Conclusion

This research provides a phonetically grounded diagnostic of SER failure under noise. The fundamental vulnerability lies in the phonetic asymmetry between robust activation cues (F0, energy) and fragile valence cues (spectral dynamics, voice quality). Standard feature sets and classifiers exploit this asymmetry by defaulting to easy-to-learn but noise-vulnerable energy-based patterns. Future work must develop architectures explicitly designed to preserve and leverage the subtle phonetic signatures that encode emotional valence, the true challenge in robust, real-world emotion recognition.